# 15-663 Computational Photography, Fall 2026

Assignment 4 Report

Manan Shah (mshah2)

## 1 Lightfield rendering

Splitting the $6400 \times 11200$ RGB image into $16 \times 16$ lenslet blocks gives $L(u, v, s, t, c)$ with shape $(16, 16, 400, 700, 3)$. I reverse the horizontal angular index to give both aperture axes the same orientation relative to their image axes. Aperture coordinates run from $-7.5$ to 7.5.

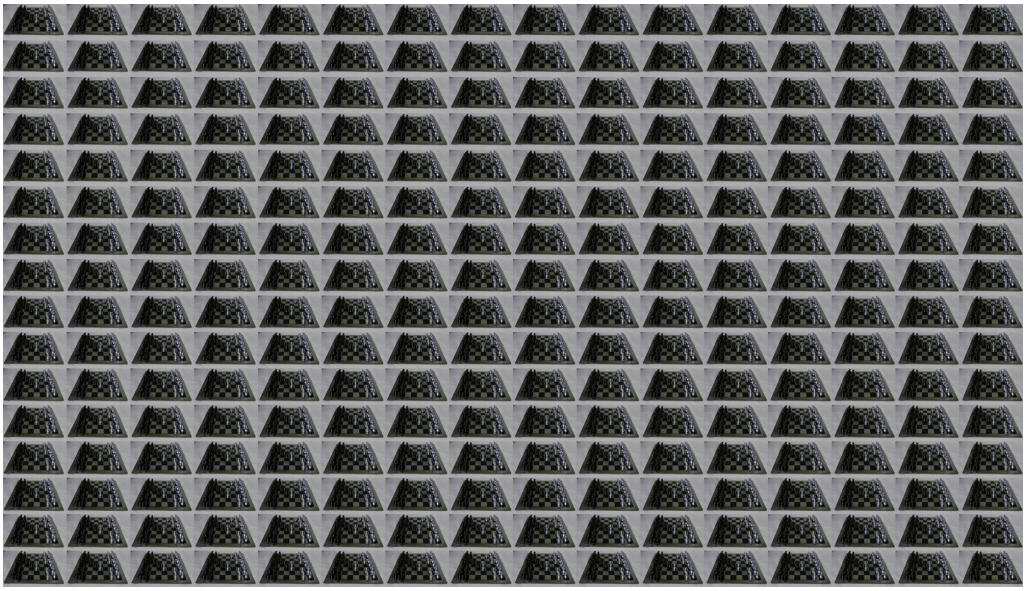


Figure 1: All 256 sub-aperture views. Aperture $u$ increases downward and $v$ increases to the right.

For refocusing, I bilinearly sample each view at $(s + du, t + dv)$ and average all 256 views in linear RGB. I use 51 depths from $-0.5$ to 2 in steps of 0.05; larger $d$ focuses nearer the front of the board. A common 16-pixel border crop removes samples outside the image, leaving $368 \times 668$ pixels. Color results are sRGB-encoded for display.




Figure 2: Five refocusings at the largest aperture.

## 2 All-in-focus image and depth from focus

I extract linear XYZ luminance $Y$, compute $H = Y - G_{\sigma_1} * Y$, then use $w = G_{\sigma_2} * H^2$ as the sharpness weight. Weighted averages of RGB values and focus parameters give the all-in-focus image and depth map, respectively.

I compare $(\sigma_1, \sigma_2) = (0.5, 1), (1, 2), (1, 4), (2, 4)$ in pixels. I choose $(1, 4)$: it reduces speckle compared with the smaller windows without the additional broadening from $\sigma_1 = 2$.

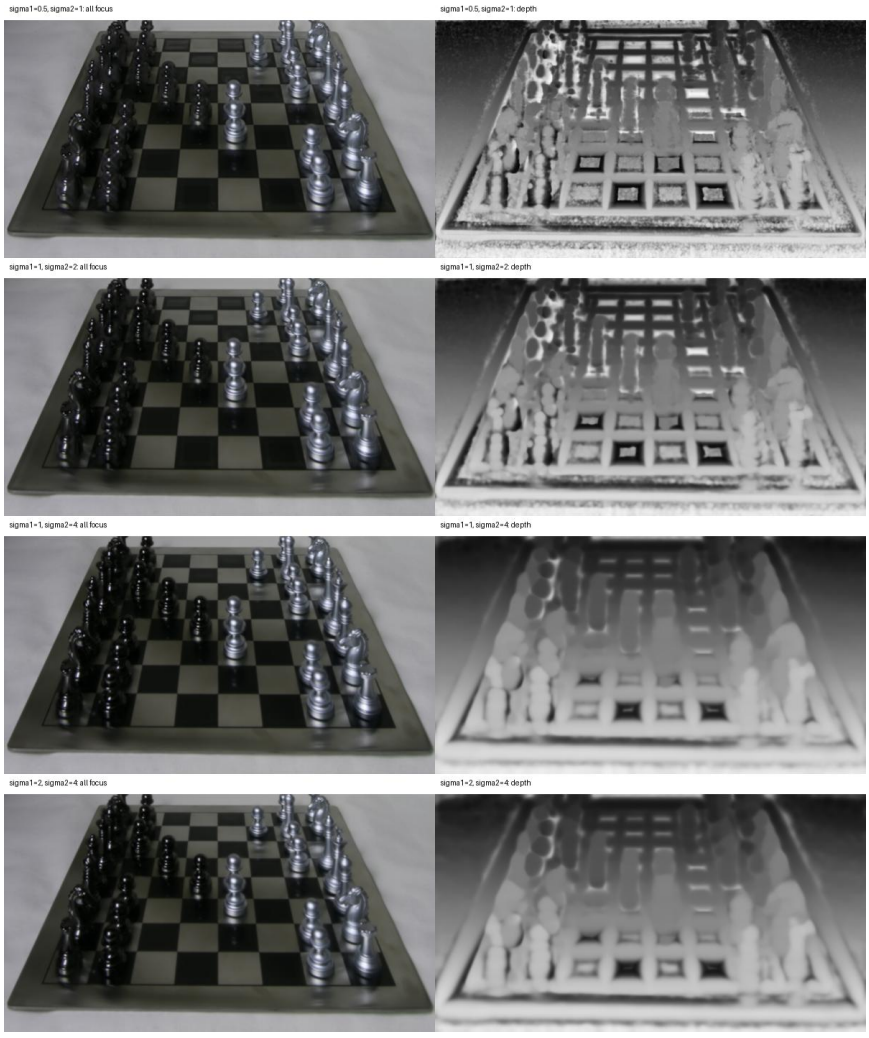



Figure 3: The four settings above, in order. Left: all-in-focus image. Right: depth; black is $d = -0.5$ and white is $d = 2$.

Depth is unreliable inside flat board squares and on the smooth background, where sharpness provides little depth information. Reflections on the silver pieces and depth boundaries also give inconsistent weights. Spatial averaging spreads strong edges into neighboring pixels, producing board-pattern artifacts and softened silhouettes. The color image is less affected in flat regions: their colors are similar at different focus settings, so incorrect depth need not change the blended color. Boundaries can still soften or form halos.

## 3 Focal-aperture stack and confocal stereo

I use square apertures with side lengths $2, 4, 8, 12, 16$, containing $4, 16, 64, 144, 256$ views. Each image is already divided by its view count, so no second exitance normalization is applied. Following Equations (7)–(8) of Hasinoff and Kutulakos, *Confocal Stereo* (ECCV 2006), I choose the depth with minimum variance across aperture, averaging the three linear RGB channel variances.

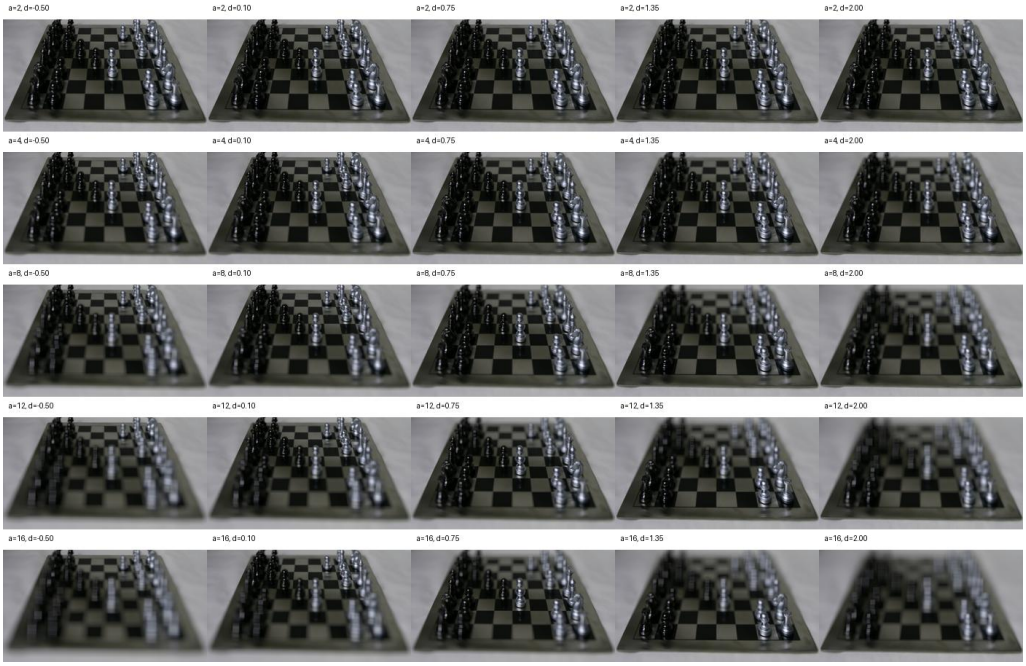
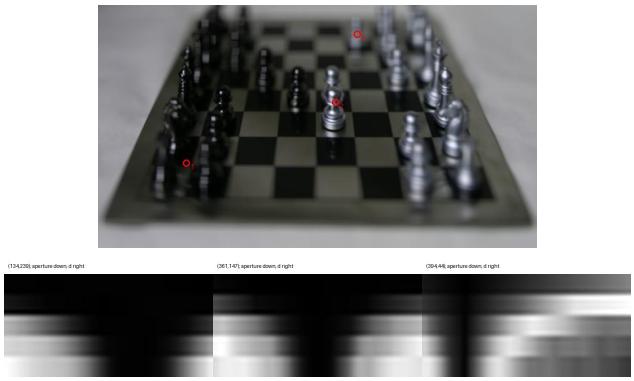



Figure 4: Aperture increases downward and focus parameter increases to the right. All tiles are exposure-normalized.



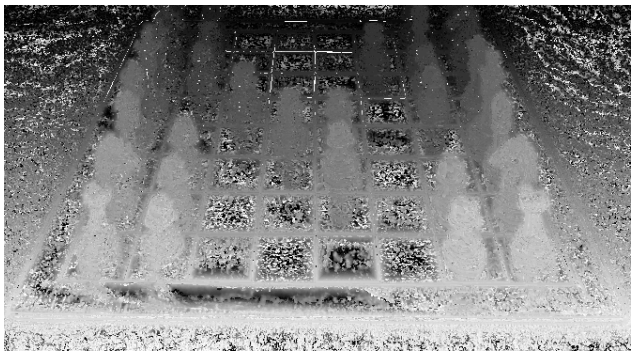


Figure 5: Left: sample locations and their three AFIs, ordered by location number. In each AFI, aperture increases downward and focus increases rightward; luminance is independently rescaled for visibility. Right: confocal depth on the same black-to-white scale as the depth-from-focus maps.

Both depth methods recover the broad back-to-front trend. Confocal stereo preserves some sharper piece boundaries but is much noisier on the background and inside board squares. Flat regions have several nearly equal variance minima, making their depth unstable. Reflections and occlusion also violate aperture constancy. Depth from focus is smoother because it averages spatial neighborhoods.

# 4 Own lightfield

I captured a phone video while translating sideways and vertically across a wallet, notebook and bottle. I used 6.93 seconds at $1080 \times 1920$, resized to $540 \times 960$, and processed every third frame (70 frames).

I selected $61 \times 61$ grayscale templates in the middle sampled frame on the wallet label, notebook lettering and bottle lettering. The search extends up to 200 pixels in each direction, clipped to the frame.

To implement Equation (9), I subtract the template mean to obtain $g_0$. Correlations with an all-ones kernel give the local pixel sum $S$ and squared sum $Q$ of a grayscale frame $I$. NCC is

$$h = \frac{I \star g_0}{\sqrt{(\sum g_0^2)(Q - S^2/n)}},$$

where $n$ is the template pixel count. Since $\sum g_0 = 0$, subtracting the local image mean from the numerator is unnecessary. I evaluate these terms using pixel-wise operations and SciPy correlations.

For each frame, I find the NCC maximum at $(x_t, y_t)$ and translate by $(x_0 - x_t, y_0 - y_t)$, where $(x_0, y_0)$ is the reference template's top-left corner. I average the aligned frames in linear RGB and crop to their common overlap.

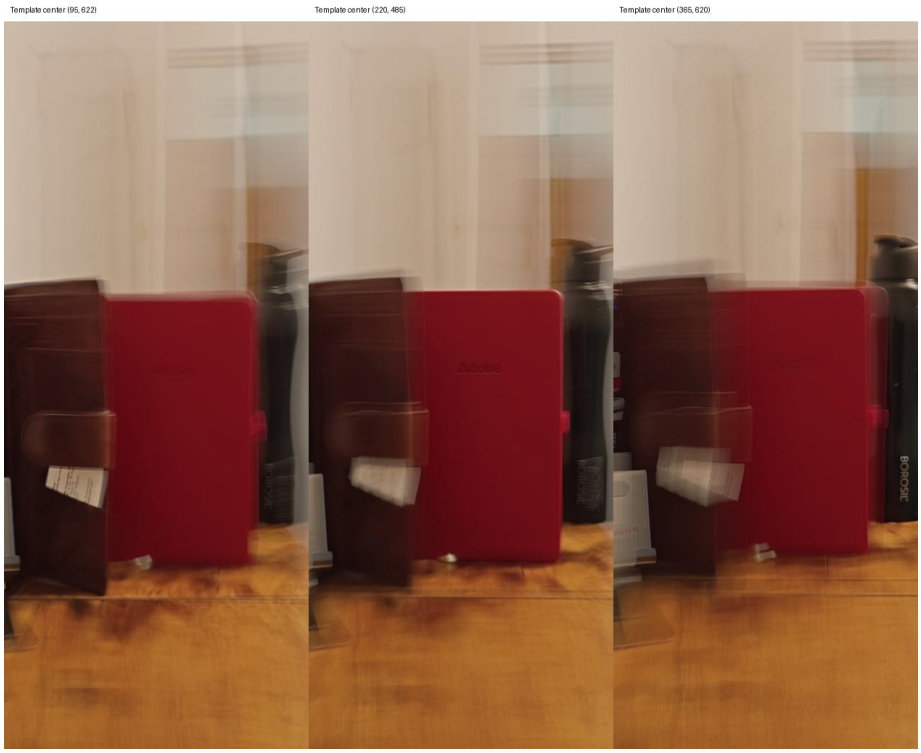



Figure 6: Refocusing on the wallet, notebook and bottle, respectively. All results share a $369 \times 880$ crop.

The selected lettering is clearest in its corresponding result. Other depths blur from parallax. The background blur is streaked because of the finite, uneven camera path; small viewpoint changes also leave residual softness that translation alone cannot correct.